

Analyzing and Mitigating Demographic Biases in Transformer-Based Automated Essay Scoring

Anonymous ACL submission

Abstract

Automated Essay Scoring (AES) systems built on transformer language models risk inheriting and amplifying demographic biases in their training data, with direct consequences for students at scale. We extend Kwako and Ormerod (2024) in two directions. First, we replicate their bias analysis across XLNet, RoBERTa, and Longformer on PERSUADE 2.0 (Crossley et al., 2024) and ASAP 2.0 (Crossley et al., 2025), finding that demographic score disparities recur across architectures. Second, we evaluate two bias-mitigation strategies—adversarial debiasing via a Gradient Reversal Layer (in-training) and orthogonal projection of hidden states (post-hoc). On XLNet, neither method reduces prediction-level bias, even though held-out demographic recoverability from the encoder is minimal to begin with ($\kappa \approx 0.2$) and projection cannot lower it further. The disparity is mediated by construct-relevant features correlated with group membership rather than carried by a removable demographic direction. Representation-level debiasing thus does not guarantee fairer scores. Code and data are available on [GitHub](#) and [HuggingFace](#).

1 Introduction

Automated Essay Scoring (AES) can be cast as a text-classification task well suited to transformer models, but these models are pretrained on corpora that reflect societal inequities and can inherit and amplify demographic biases during fine-tuning. Deployed at scale, a biased AES system can distort grades and harm the prospects of students in affected groups, especially those already underserved (Kwako and Ormerod, 2024; Hardt et al., 2016). Ensuring such systems are fair before deployment is therefore essential.

Kwako and Ormerod (2024) showed that XLNet fine-tuned on PERSUADE amplifies human score differences across race and gender and im-

plicitly encodes demographic information in its hidden states; related work extends bias findings to other languages and to LLM scoring (Schaller et al., 2024; Yamashita, 2025). Two gaps remain: existing analyses are largely confined to a single model and dataset, and while recent work proposes AES debiasing methods (Al Abdullah et al., 2026; Fan and Yun, 2026), it reports aggregate performance rather than testing whether debiasing reduces score disparities relative to the human baseline. We address both. We replicate the bias pipeline across XLNet, RoBERTa, and Longformer on PERSUADE 2.0 and ASAP 2.0, and evaluate two mitigations that target the encoded demographic signal directly: adversarial debiasing via a Gradient Reversal Layer (Zhang et al., 2018; Ganin and Lempitsky, 2015) and orthogonal projection of hidden states (Bolukbasi et al., 2016; Ravfogel et al., 2020).

2 Related Work

2.1 Automated Essay Scoring.

Transformer-based models have become the dominant approach for AES. Devlin et al. (2019) introduced BERT, the encoder-only architecture underlying most modern AES systems, while Yang et al. (2019) proposed XLNet with a generalized autoregressive objective well-suited to longer texts. Rodriguez et al. (2019) directly compared BERT and XLNet on the ASAP dataset, providing the most direct precedent for our multi-model comparison. Wang et al. (2022) further showed that standard BERT usage was suboptimal for AES and introduced multi-scale essay representations to address this. More recently, Ormerod and Kehat (2025) benchmarked DeBERTa, XLNet, Longformer, and Mamba on ASAP 2.0, which we treat as a direct baseline for our first contribution.

2.2 Demographic Bias in AES.

Kwako and Ormerod (2024) established that XL-Net fine-tuned on PERSUADE amplifies marginal human score differences across race and gender, and that demographic information is implicitly encoded in the model’s hidden states. This finding has been extended in several directions: Schaller et al. (2024) examined fairness across shallow, deep, and LLM-based approaches on German-language essays; Yang et al. (2025) investigated whether zero-shot and few-shot LLMs exhibit similar biases; Yamashita (2025) analyzed GPT-4o’s scores on the ELLIPSE corpus using many-facet Rasch measurement; and Andersen et al. (2025) studied fairness in short-answer scoring under the EU AI Act’s high-risk classification framework.

2.3 Bias Mitigation.

Zhang et al. (2018) established the foundational adversarial debiasing framework, and Han et al. (2021) adapted it for NLP tasks, showing that representational fairness does not always imply empirical fairness. Bolukbasi et al. (2016) introduced geometric debiasing by projecting embeddings onto the subspace orthogonal to a bias direction. This is the mathematical basis of our second mitigation approach. Orgad and Belinkov (2023) demonstrated that bias removal is possible even without demographic annotations at training time using a DeBERTa backbone, which is practically relevant given incomplete demographic coverage in some PERSUADE prompts. Most directly related to our work, Al Abdullah et al. (2026) proposes FairGrade, a BERT-based adversarial model with a Gradient Reversal Layer that improves QWK on ASAP, and Fan and Yun (2026) applies contrastive learning on DeBERTa-v3 to reduce bias against ESL learners on ASAP 2.0 and ELLIPSE, both of which serve as baselines for our Contribution 2.

3 Datasets

3.1 PERSUADE 2.0.

The PERSUADE 2.0 corpus (Crossley et al., 2024) is a large-scale collection of argumentative and persuasive student essays produced by 6th-12th grade students in the United States. It contains over 25,000 essays across 15 prompts spanning two writing tasks: independent and source-based writing. The dataset includes rich demographic and individual metadata for each writer, as well as holistic essay scores and proficiency scores for

each argumentative and discourse element. We use the provided train/test split of 15,594 and 10,402 essays respectively.

3.2 ASAP 2.0.

The ASAP 2.0 benchmark (Crossley et al., 2025) comprises 24,278 source-based persuasive essays collected from standardized writing tests at the state level, written by 6th, 8th, 9th, and 10th grade students across seven prompts. Of these, 12,871 essays are sourced from the PERSUADE 2.0 corpus and 11,829 are previously unreleased. Essays were selected to maximize demographic coverage, including English Language Learning (ELL) status, economic background, disability status, race/ethnicity, gender, and grade level. We use the provided train/test split of 17,307 and 7,421 essays respectively.

3.3 Evaluation Metrics.

Model performance is assessed using the three standard AES metrics (QWK, SMD, and exact agreement) as defined by Williamson et al. (2012). We use the standardized mean difference (SMD) in two distinct senses, which we keep separate throughout: as an *accuracy* check it is the standardized machine-human mean score difference for a model, while as a *disparity* measure it is the standardized between-group difference in scores; the latter, expressed as the weighted regression z below, is our bias metric.

3.4 Demographic Score Distributions

Because we measure model bias in relative terms – that is, whether automated scores amplify or attenuate differences already present in human scores – it is necessary to first characterize the human-score baseline. Table 1 reports pairwise standardized mean differences (SMD) between demographic groups for human-assigned scores in both corpora, computed as Cohen’s d using pooled standard deviations. All differences are statistically significant ($p < 0.001$).

Attribute	ASAP SMD	PERSUADE SMD
Disability status	0.644 ^m	0.566 ^m
ELL status	0.535 ^m	0.626 ^m
Economic disadvantage	0.375 ^s	0.575 ^m
Gender (F–M)	0.209 ^s	0.198 ⁿ

Table 1: Human-score SMDs by demographic group. Superscripts: ⁿ=negligible, ^s=small, ^m=medium (Cohen, 1988). All $p < 0.001$.

Several patterns warrant attention. First, disability status and ELL status produce the largest gaps in both corpora, with medium effect sizes that exceed the gender gap which has received more attention in prior AES bias work. Second, for race and ethnicity, scores follow a consistent gradient across both datasets: Asian/Pacific Islander students score highest, followed by White, Hispanic/Latino, and Black/African American students. The Black–White gap is 0.43 points in ASAP 2.0 and 0.29 points in PERSUADE 2.0. Third, disadvantage compounds across groups: in PERSUADE 2.0, economically disadvantaged Black students achieve a mean score of 2.95, compared to 4.20 for non-disadvantaged Asian students – a gap of 1.25 points driven by intersecting demographic factors rather than any single attribute. Finally, approximately 24% of essays are missing at least one demographic field in both datasets, and this missingness is statistically non-random ($p < 0.001$), requiring care in downstream analyses.

These human-score baselines serve as the reference point for all model bias evaluations reported in Section 5: a model *amplifies* bias when its predicted score differences exceed the corresponding human baseline, and *attenuates* bias when they fall below it.

4 Methodology

4.1 Data Preparation & Tokenization

With PERSUADE 2.0 and ASAP 2.0, we now have access to demographic information such as Gender, Race/Ethnicity, English Language Learner (ELL) status, and Economically Disadvantaged (SES) status, and Disability status of each student, albeit full information for each student is not guaranteed. Regarding rows with missing scores, we decided to drop them to make sure PyTorch does not run into collation errors during training, developing, and testing. We also followed the train, test split that the original Kaggle competition had. We also followed the approach that Kwako and Ormerod (2024) implemented to make sure that our re-implementation of their approach is preserved for apple to apple comparison. Because we are benchmarking performance between 3 different models, we decided to implement a modular tokenization approach that can support different architecture.. Before feeding data into the tokenizer, we made sure to standardized our dataset with *truncation = True* and setting *max_length = 2048* to handle long essays

and preventing memory crashes.

4.2 Model Architecture & Fine-Tuning

We first re-implement XLNet to serve as the baseline, and then introduce RoBERTa and Longformer. All three models were loaded with its main body and a final regression layer head using Mean Squared Error (MSE) loss, to ensure the ordinal ranking of prediction, to output the predicted essay score. AdamW optimizer was employed with the following hyperparameters:

Table 2: Model Hyperparameter

learning rate	5×10^{-6}
batch size	8
maximum epochs	20

We also implement early stopping while examining model MSE on dev set after every epochs to make sure that overfitting does not occur.

4.3 Automated Bias Evaluation

Following Kwako and Ormerod (2024), we measure demographic scoring bias in relative terms – that is, we ask whether each model amplifies or attenuates human score differences between demographic groups, rather than whether group differences exist in absolute terms. This framing avoids conflating model-introduced bias with pre-existing, construct-relevant differences in writing proficiency across groups.

To adjust for marginal group differences in human-assigned scores (e.g., female students on average score higher than male students on persuasive writing), we apply exact matching (Kwako and Ormerod, 2024). Specifically, we compare students within the same human score point across demographic groups – male and female students who both received a 2 are compared to each other, and so on – producing a set of sample weights that eliminate first-order group differences. These weights are used in all subsequent bias analyses.

We estimate pairwise group differences between model- and human-assigned scores with weighted regression on the exact-matched samples, clustering standard errors at the prompt level and applying the Benjamini–Hochberg procedure to control the false discovery rate across comparisons. Differences are reported as z -scores so effect sizes are comparable across prompts. Score differences are normalized to z -scores so that effect sizes are comparable across prompts and corpora. We evaluate

bias across all available demographic axes in each corpus: race/ethnicity, gender, ELL status, SES status, and disability status for PERSUADE 2.0; race/ethnicity, gender, and ELL status for ASAP 2.0. This pipeline is applied identically to all three models (XLNet, RoBERTa, Longformer), enabling a direct cross-architecture comparison of bias patterns.

4.4 Embedded Features Extraction

To determine whether demographic information is implicitly encoded within each scoring model’s learned representations, we follow the probing methodology of Kwako and Ormerod (2024). Each fine-tuned scoring model $\mathcal{M}$ can be decomposed as:

$$\mathcal{M}(x) = \underbrace{(\sigma \circ \mathcal{L})}_{\text{Classifier}} \circ \underbrace{(\mathcal{S} \circ \mathcal{T} \circ \mathcal{E})}_{\text{feature model}}(x) \quad (1)$$

where $\mathcal{E}$ is the token embedding layer, $\mathcal{T}$ is the stack of transformer layers, $\mathcal{S}$ is a summary layer, $\mathcal{L}$ is the regression head, and σ is the activation function. The composition $(\mathcal{S} \circ \mathcal{T} \circ \mathcal{E})$ constitutes the *feature model* – the portion that maps input text to a fixed-dimensional hidden representation – which is subsequently used by the linear head to predict a score.

To test whether demographic information is embedded in this representation, we freeze the feature model after fine-tuning and replace the regression head with a classification head $\tilde{\mathcal{L}}$, optimizing only $\tilde{\mathcal{L}}$ to predict each demographic attribute in turn (gender, race/ethnicity, ELL status, SES status, disability status):

$$\tilde{\mathcal{M}}(x) = (\sigma \circ \tilde{\mathcal{L}}) \circ (\mathcal{S} \circ \mathcal{T} \circ \mathcal{E})(x) \quad (2)$$

If $\tilde{\mathcal{M}}$ can classify demographic affiliation above chance, we conclude that the attribute is *implicit* in the model’s hidden representation. Classification performance is reported using Cohen’s κ (Kwako and Ormerod, 2024), where $\kappa < 0.2$ indicates near-chance performance and $\kappa \geq 0.4$ indicates meaningful encoding of demographic information. This analysis is applied identically to all three model architectures, allowing us to compare not only whether demographic features are encoded, but whether the degree of encoding differs across architectures and correlates with the scoring biases observed in Section 4.3.

Prompt	Ours	K&O 2024
“A Cowboy Who Rode the Waves”	0.697	0.755
Car-free cities	0.757	0.796
Cell phones at school	0.705	0.800
Community service	0.707	0.800
Distance learning	0.848	0.875
Does the electoral college work?	0.853	0.863
Driverless cars	0.710	0.774
Exploring Venus	0.806	0.856
Facial action coding system	0.856	0.865
Grades for extracurricular act.	0.693	0.779
Mandatory extracurricular act.	0.810	0.815
Phones and driving	0.735	0.781
Seeking multiple opinions	0.600	0.674
Summer projects	0.764	0.813
The Face on Mars	0.783	0.818
Macro Avg.	0.755	0.864

Table 3: Per-prompt QWK: our XLNet replication vs. Kwako and Ormerod (2024).

5 Evaluation & Results

5.1 XLNet Replication

We fine-tune xlnet-base-cased per prompt on PERSUADE 2.0 with K&O’s hyperparameters (Table 3). Our macro QWK of 0.755 is 0.109 below their 0.864, consistent with the shift from PERSUADE 1.0 (their training corpus) to 2.0.

We report bias as the weighted cluster-robust regression z of Kwako and Ormerod (2024), comparable to their estimates and to our mitigation results (Section 6.3).¹ For gender, our baseline yields $z = +0.146$ (per-prompt mean; $+0.095$ pooled) against K&O’s $+0.06$, with a closely matching coefficient ($+0.049$, $SE = 0.011$): a faithful reproduction of their small female-over-male advantage, slightly amplified. Bias is far larger for the other attributes—ELL $z = +1.730$ (an order of magnitude above gender), disability $+0.597$, economic disadvantage $+0.476$ —and in every case the model amplifies the human-score gap (Table 1) rather than attenuating it. ELL, both the largest disparity and the attribute Kwako and Ormerod (2024) found most strongly encoded, is the most demanding test for the mitigations in Section 6.

5.2 RoBERTa Implementation

We implemented the same steps but on RoBERTa to examine cross-model effects. RoBERTa only allows maximum of 512-token per input which leads to truncation of some long essay response.

¹An earlier framing compared a pooled Cohen’s d to K&O’s regression z ; as different estimators, that overstated the divergence. All bias numbers here use the regression z .

5.2.1 AES Performance

Despite the limit, RoBERTa still perform relatively well in AES with QWK > 0.70 as per the standard industry on both corpus, indicating high human grading mimicking capability even with truncated essay input.

Table 4: Test Set Performance for RoBERTa

Corpus	QWK	SMD	Accuracy
PERSUADE	0.8370	0.3613	0.6152
ASAP	0.7991	0.1043	0.6612

5.2.2 Structural Demographic Bias

However, RoBERTa seems to amplify more bias comparing to XLNet baseline. In PERSUADE corpus, RoBERTa heavily favored Asian/Pacific Islander students over Black/African American students with $Z = 5.273$ (statistically significant as well). This amplification can also be seen in disparity amongst grade given to English Language Learners (ELL) versus non-ELL students ($Z = -5.958, p < 0.001$), as well as with studnets identified as having a disability ($Z = -2.529, p < 0.05$).

We took a step further than just cross-model comparison, and perform analysis on ASAP corpus as well. RoBERTa, on ASAP, maintained its bias amplification on ELL students ($Z = -2.644, p < 0.05$) and students with disabilities ($Z = -8.128, p < 0.001$). Interestingly, students who are not economically disadvantaged was favored by RoBERTa ($Z = 3.875, p < 0.001$).

5.3 Longformer Implementation

We fine-tune allenai/longformer-base-4096 on both PERSUADE 2.0 and ASAP 2.0 using the same regression head and MSE loss setup as the XLNet baseline, with AdamW at learning rate 5×10^{-6} , batch size 4, and early stopping on dev MSE. Due to GPU memory constraints on the V100, maximum token length was reduced to 1,024.

Table 5 reports performance on both corpora. On PERSUADE 2.0, Longformer achieves a QWK of 0.851, exact agreement of 0.659, and SMD of 0.136, stopping at epoch 4. On ASAP 2.0, Longformer achieves a QWK of 0.809, exact agreement of 0.665, and SMD of 0.028, stopping at epoch 7. Both results exceed the operational standard of $QWK \geq 0.7$. The PERSUADE SMD of 0.136 falls just below the commonly accepted threshold of

0.15, while the ASAP SMD of 0.028 is well within range.

Corpus	QWK	SMD	Acc
PERSUADE 2.0	0.851	0.136	0.659
ASAP 2.0	0.809	0.028	0.665

Table 5: Longformer performance on PERSUADE 2.0 and ASAP 2.0.

6 Bias Mitigation

Our replication establishes that all three architectures amplify human score gaps, while demographic attributes are only weakly recoverable from each model’s hidden states by a held-out linear probe ($\kappa \approx 0.2$; Section 4.3). This motivates the question Kwako and Ormerod (2024) leave open: can this encoding be removed without sacrificing scoring quality? We evaluate two interventions that target it directly—adversarial debiasing via a Gradient Reversal Layer (in-training) and orthogonal projection of hidden states (post-hoc)—chosen to span the in-training/post-hoc design space.

6.1 Adversarial Debiasing via Gradient Reversal Layer

We augment the scoring model with an adversarial classifier f_d predicting the protected attribute from the same final-token hidden state h used by the scoring head f_s . Following Ganin and Lempitsky (2015), a Gradient Reversal Layer is the identity on the forward pass and negates the gradient on the backward pass, so the encoder receives

$$\nabla_{\theta_E} \mathcal{L}_s - \lambda \cdot \nabla_{\theta_E} \mathcal{L}_d, \quad (3)$$

with $\mathcal{L}_s$ the scoring (MSE) and $\mathcal{L}_d$ the adversarial (cross-entropy) loss; at equilibrium the adversary is at chance and h carries no linearly recoverable demographic signal (Elazar and Goldberg, 2018). We use a two-layer MLP adversary (hidden dim 128) and ramp λ from 0 to $\lambda_{\max}=0.5$ ($\gamma=10$) to avoid early instability (Ganin and Lempitsky, 2015). For imbalanced attributes (ELL $\approx 9\%$, disability $\approx 10\%$) we weight $\mathcal{L}_d$ inversely to class frequency, without which the adversary collapses to the majority class (He and Garcia, 2009). Each prompt is trained independently and the adversary discarded after training.

6.2 Orthogonal Projection on Demographic Subspaces

In contrast to GRL, orthogonal projection (Bolukbasi et al., 2016; Ravfogel et al., 2020) operates on a frozen model. We estimate the demographic direction $w_d \in R^{768}$ from a linear probe $f_d(h) = \sigma(w_d^\top h + b)$ on the frozen encoder (Section 4.3), form the projector

$$P = I - \frac{w_d w_d^\top}{\|w_d\|^2}, \quad (4)$$

and score from the projected state $\hat{y} = f_s(Ph)$, realizing the §4.3 framing of Kwako and Ormerod (2024). Since a single w_d need not capture all recoverable signal, we also apply Iterative Null-space Projection (Ravfogel et al., 2020), re-probing and re-projecting until a fresh probe is at chance; we report both variants. Projection requires no retraining but assumes the signal lies in a low-dimensional linear subspace.

6.3 Mitigation Results

We evaluate both interventions on XLNet/PERSUADE 2.0, tracking encoder recoverability (held-out probe κ), prediction bias (weighted regression z), and scoring quality (macro QWK). The finding is a dissociation: held-out demographic encoding is minimal throughout and the interventions barely change it, yet they leave prediction bias essentially unchanged (Table 7).

Adversarial debiasing (GRL). With $\lambda_{\max} = 0.5$, GRL drives the adversary to chance while preserving scoring quality (macro QWK 0.766 vs. 0.755), but the weighted z does not fall: gender is indistinguishable from baseline ($+0.146 \rightarrow +0.167$) and ELL—the largest baseline bias—moves only $+1.730 \rightarrow +1.640$. For imbalanced attributes the adversary oscillates between all-one-class solutions, so we judge GRL by prediction-side metrics, not adversary accuracy.

Orthogonal projection. Held-out recoverability is minimal even before projection ($\kappa \approx 0.18 - 0.26$), and a single step lowers it only marginally (ELL $0.22 \rightarrow 0.20$, disability $0.18 \rightarrow 0.16$) while macro QWK is essentially unchanged (≤ 0.003). An in-sample probe suggests a far larger drop ($\kappa \approx 0.9 \rightarrow 0.5$), but this does not generalize: the projection removes directions that fit the probe’s training split, not a stable demographic subspace.

Attribute	Length only	All features
Gender	+68%	+99%
ELL	−66%	−118%
Economic	−29%	−18%
Disability	−51%	−57%

Table 6: Proportion of the demographic score gap mediated by surface features (word count alone vs. all five). Positive = explained by features; negative = *suppression* (gap widens once features are held constant). Signs stable across both specifications.

Five iterations of null-space projection reduce held-out κ by at most ~ 0.05 and still leave z large (ELL +1.40 on QWK-preserved prompts, disability +0.55). The prediction-level z moves by at most 0.05 on every attribute (Table 7).

What the residual bias is. Regressing predicted score on group membership before and after controlling for surface features (length, sentence count, word length, Flesch–Kincaid, lexical diversity) on the exact-matched sample (Table 6) splits the attributes sharply. The gender gap is almost entirely mediated by these features (99%; 68% by length alone): at equal surface quality XLNet shows essentially no gender disparity, matching the length-mediation Kwako and Ormerod (2024) report. For ELL, economic, and disability status the pattern reverses—controlling for features *increases* the gap (e.g. ELL standardized $-0.135 \rightarrow -0.294$), indicating the model penalizes these groups beyond what their writing features predict. This explains the mitigation failure on both sides: for gender the gap rides on length, a score-predictive feature the head retains after the demographic axis is removed; for the others it is a residual penalty distributed beyond any single linear direction. Neither failure mode is addressed by erasing a demographic axis, which is why both GRL and projection leave the disparities intact. This is the empirical realization of the warning (Elazar and Goldberg, 2018; Ravfogel et al., 2020) that representational guarding need not yield behavioral fairness—and it holds here for a sharper reason: held-out recoverability was minimal at the outset (gender length→attribute $\kappa = 0.106$ in Kwako and Ormerod, 2024), so there was little linear signal to remove, yet the prediction gap persists.

7 Limitations and Future Work

Our bias estimates are measured relative to human-assigned scores, which may themselves be biased;

Method	Attr.	z pre	z post	κ
GRL	gender	+0.15	+0.17	adv→ch.
GRL	ELL	+1.73	+1.64	adv→ch.
Proj-1	ELL	+1.73	+1.68	.22→.20
Proj-1	disab.	+0.60	+0.60	.18→.16
INLP-5	ELL [†]	+1.73	+1.40	.22→.17
INLP-5	disab.	+0.60	+0.55	.18→.16

Table 7: Mitigation on XLNet/PERSUADE 2.0. z : weighted regression effect size (per-prompt mean). κ : held-out probe recoverability, before→after (“adv→ch.” = adversary at chance). [†]mean over QWK-preserved prompts. Proj-1 gender/econ show the same null pattern (App.); both z move ≤ 0.01 .

without an unbiased anchor (e.g. a DIF reference set, unavailable for these corpora) we measure amplification or attenuation of human scoring patterns, not ground-truth fairness. Approximately 24% of essays are missing at least one demographic field, and this missingness is non-random (Section 3.4); we do not correct for it via inverse-probability weighting or imputation, so estimates for underrepresented groups should be read with caution. Our mitigation experiments are confined to XLNet/PERSUADE 2.0: the replication shows bias recurs across architectures, but we do not evaluate mitigation on RoBERTa, Longformer, or ASAP 2.0, so mitigation claims are specific to XLNet. RoBERTa’s 512-token limit, far below XLNet’s 2,048, forces truncation of long essays and may affect both its scoring and the demographic signal available to it.

Held-out probing recovers demographic affiliation only weakly for most attributes ($\kappa \approx 0.2$), the clearest exception being ELL status on the *Distance learning* prompt ($\kappa = 0.78$), which reproduces the one substantial probing result of Kwako and Ormerod (2024). Even where an attribute is recoverable, correlation does not establish causation: we cannot conclude the model uses these features when scoring.

Our negative result points to clearer next steps than linear erasure. Because the disparity is mediated by construct-relevant features rather than a removable demographic direction, methods that target behavioral fairness directly—fairness-aware training objectives such as equal-opportunity losses or group DRO—and non-linear representational removal are more promising. Data-centric strategies such as targeted augmentation for ELL writers, and multi-trait scoring that models rubric dimensions explicitly rather than suppressing protected signal, are also natural directions.

8 Ethics

This work uses student essay data from minors in K–12 settings. Both PERSUADE 2.0 and ASAP 2.0 are publicly released under their respective licenses (CC BY-NC-SA 4.0 and CC BY 4.0); we use them solely for academic research in accordance with those terms, and conducted no additional data collection or human-subjects interaction.

Our central finding carries a deployment risk worth stating directly: standard representation-level fairness checks (probe recoverability, adversary accuracy) can indicate that demographic information has been removed while prediction-level disparities remain essentially unchanged. A system certified fair on representational metrics alone could therefore still disadvantage ELL, disability, and economically disadvantaged students at the score level. Fairness evaluation in deployed AES should be conducted on predicted scores relative to a defensible baseline, not on representational proxies.

9 Conclusion

We extended the bias analysis of Kwako and Ormerod (2024) across XLNet, RoBERTa, and Longformer on PERSUADE 2.0 and ASAP 2.0. We find that demographic scoring disparities are systemic across all three architectures, not artifacts of any single model, while demographic attributes are only weakly recoverable from each model’s hidden representations by a held-out probe.

Both adversarial debiasing and orthogonal projection target the encoded demographic signal, yet neither reduces prediction bias: held-out recoverability is minimal to begin with ($\kappa \approx 0.2$) and removing it—even with five iterations of null-space projection—leaves the disparity essentially unchanged across all four attributes. The gap is mediated by construct-relevant features correlated with group membership, not a removable demographic direction. Representation-level debiasing thus does not guarantee prediction-level fairness in AES, and representational metrics alone can overstate the fairness achieved.

10 Acknowledgments

References

Rowaidah Al Abdullah, Fatma Al Shamli, Amal Al Abri, Sara Al Moqbali, and Nasim Al Balushi. 2026. *Adversarial debiasing for fair automated essay scor-*

589	ing across linguistic backgrounds. <i>Scientific Culture</i> ,	645
590	12(4):1474–1489.	646
591	Nico Andersen, Julia Mang, Frank Goldhammer, and	647
592	Fabian Zehner. 2025. Algorithmic fairness in auto-	648
593	matic short answer scoring . <i>International Journal of</i>	649
594	<i>Artificial Intelligence in Education</i> , 35:3128–3165.	650
595	Tolga Bolukbasi, Kai-Wei Chang, James Zou,	651
596	Venkatesh Saligrama, and Adam Kalai. 2016. Man	652
597	is to computer programmer as woman is to home-	653
598	maker? Debiasing word embeddings. In <i>Advances in</i>	654
599	<i>Neural Information Processing Systems</i> , volume 29.	
600	Jacob Cohen. 1988. <i>Statistical Power Analysis for the</i>	
601	<i>Behavioral Sciences</i> , 2nd edition. Lawrence Erlbaum	
602	Associates.	
603	Scott A. Crossley, Perpetual Baffour, L. Burleigh, and	
604	Jules King. 2025. A large-scale corpus for assessing	
605	source-based writing quality: ASAP 2.0 . <i>Assessing</i>	
606	<i>Writing</i> , 65:100954.	
607	Scott A. Crossley, Yiyang Tian, Perpetual Baffour,	
608	Ann Franklin, Margaret Benner, and Ulrich Boser.	
609	2024. A large-scale corpus for assessing written ar-	
610	gumentation: PERSUADE 2.0 . <i>Assessing Writing</i> ,	
611	61:100865.	
612	Jacob Devlin, Ming-Wei Chang, Kenton Lee, and	
613	Kristina Toutanova. 2019. BERT: Pre-training of	
614	deep bidirectional transformers for language under-	
615	standing . In <i>Proceedings of the 2019 Conference of</i>	
616	<i>the North American Chapter of the Association for</i>	
617	<i>Computational Linguistics: Human Language Tech-</i>	
618	<i>nologies, Volume 1 (Long and Short Papers)</i> , pages	
619	4171–4186, Minneapolis, Minnesota. Association for	
620	Computational Linguistics.	
621	Yanai Elazar and Yoav Goldberg. 2018. Adversarial	
622	removal of demographic attributes from text data .	
623	In <i>Proceedings of the 2018 Conference on Empiri-</i>	
624	<i>cal Methods in Natural Language Processing</i> , pages	
625	11–21, Brussels, Belgium. Association for Computa-	
626	tional Linguistics.	
627	Kevin Fan and Eric Yun. 2026. Mitigating bias	
628	in automated grading systems for ESL learners:	
629	A contrastive learning approach . <i>arXiv preprint</i>	
630	<i>arXiv:2601.16724</i> .	
631	Yaroslav Ganin and Victor Lempitsky. 2015. Unsu-	
632	ervised domain adaptation by backpropagation. In	
633	<i>Proceedings of the 32nd International Conference</i>	
634	<i>on Machine Learning</i> , volume 37 of <i>Proceedings of</i>	
635	<i>Machine Learning Research</i> , pages 1180–1189.	
636	Xudong Han, Timothy Baldwin, and Trevor Cohn. 2021.	
637	Decoupling adversarial training for fair NLP . In	
638	<i>Findings of the Association for Computational Lin-</i>	
639	<i>guistics: ACL-IJCNLP 2021</i> , pages 471–477, Online.	
640	Association for Computational Linguistics.	
641	Moritz Hardt, Eric Price, and Nathan Srebro. 2016.	
642	Equality of opportunity in supervised learning. In	
643	<i>Advances in Neural Information Processing Systems</i> ,	
644	volume 29.	
	Haibo He and Edwardo A. Garcia. 2009. Learning from	645
	imbalanced data . <i>IEEE Transactions on Knowledge</i>	646
	<i>and Data Engineering</i> , 21(9):1263–1284.	647
	Alexander Kwako and Christopher Ormerod. 2024. Can	648
	language models guess your identity? Analyzing de-	649
	mographic biases in AI essay scoring . In <i>Proceed-</i>	650
	<i>ings of the 19th Workshop on Innovative Use of NLP</i>	651
	<i>for Building Educational Applications (BEA 2024)</i> ,	652
	pages 78–86, Mexico City, Mexico. Association for	653
	Computational Linguistics.	654
	Hadas Orgad and Yonatan Belinkov. 2023. BLIND:	655
	Bias removal with no demographics . In <i>Proceedings</i>	656
	<i>of the 61st Annual Meeting of the Association for</i>	657
	<i>Computational Linguistics (Volume 1: Long Papers)</i> ,	658
	pages 8801–8821, Toronto, Canada. Association for	659
	Computational Linguistics.	660
	Christopher Ormerod and Gitit Kehat. 2025. Long con-	661
	text automated essay scoring with language models.	662
	In <i>Proceedings of the 20th Workshop on Innovative</i>	663
	<i>Use of NLP for Building Educational Applications</i>	664
	<i>(BEA 2025)</i> .	665
	Shauli Ravfogel, Yanai Elazar, Hila Gonen, Michael	666
	Twiton, and Yoav Goldberg. 2020. Null it out: Guard-	667
	ing protected attributes by iterative nullspace projec-	668
	tion . In <i>Proceedings of the 58th Annual Meeting of</i>	669
	<i>the Association for Computational Linguistics</i> , pages	670
	7237–7256, Online. Association for Computational	671
	Linguistics.	672
	Pedro Uriá Rodríguez, Amir Jafari, and Christopher M.	673
	Ormerod. 2019. Language models and automated	674
	essay scoring . In <i>Proceedings of the Workshop on</i>	675
	<i>Innovative Use of NLP for Building Educational Ap-</i>	676
	<i>plications</i> , pages 58–67, Florence, Italy. Association	677
	for Computational Linguistics.	678
	Nils-Jonathan Schaller, Yuning Ding, Andrea Horbach,	679
	Jennifer Meyer, and Thorben Jansen. 2024. Fairness	680
	in automated essay scoring: A comparative analy-	681
	sis of algorithms on German learner essays from	682
	secondary education . In <i>Proceedings of the 19th</i>	683
	<i>Workshop on Innovative Use of NLP for Building Ed-</i>	684
	<i>ucational Applications (BEA 2024)</i> , pages 210–221,	685
	Mexico City, Mexico. Association for Computational	686
	Linguistics.	687
	Yupei Wang, Shaohua Yang, Longtian Sun, and Qika	688
	Lin. 2022. On the use of BERT for automated essay	689
	scoring: Joint learning of multi-scale essay represen-	690
	tation . In <i>Proceedings of the 2022 Conference of</i>	691
	<i>the North American Chapter of the Association for</i>	692
	<i>Computational Linguistics: Human Language Tech-</i>	693
	<i>nologies</i> , pages 3416–3425, Seattle, United States.	694
	Association for Computational Linguistics.	695
	David M. Williamson, Xiaoming Xi, and F. Jay Breyer.	696
	2012. A framework for evaluation and use of auto-	697
	mated scoring . <i>Educational Measurement: Issues</i>	698
	<i>and Practice</i> , 31(1):2–13.	699
	Taichi Yamashita. 2025. Exploring potential biases	700
	in GPT-4o’s ratings of English language learners’	701
	essays . <i>Language Testing</i> , 42(3):344–358.	702

- 703 Kaixun Yang, Mladen Raković, Dragan Gašević, and
704 Guanliang Chen. 2025. Does the prompt-based
705 large language model recognize students’ demo-
706 graphics and introduce bias in essay scoring? In
707 *Proceedings of the International Conference on*
708 *Artificial Intelligence in Education (AIED 2025)*.
709 ArXiv:2504.21330.
- 710 Zhilin Yang, Zihang Dai, Yiming Yang, Jaime Car-
711 bonell, Russ R. Salakhutdinov, and Quoc V. Le. 2019.
712 XLNet: Generalized autoregressive pretraining for
713 language understanding. In *Advances in Neural In-*
714 *formation Processing Systems*, volume 32.
- 715 Brian Hu Zhang, Blake Lemoine, and Margaret Mitchell.
716 2018. Mitigating unwanted biases with adversarial
717 learning. In *Proceedings of the 2018 AAAI/ACM*
718 *Conference on AI, Ethics, and Society*, pages 335–
719 340.